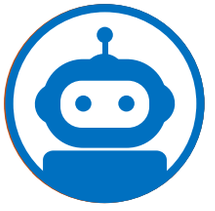
Problem Generator

"time_horizon": "90",
"demand_type": "random",
"demand_distribution": "poisson(8)",
"state_transition_model": "lost_sale",
"holding_cost": "0.5",
"penalty_cost": "2.83",
"setup_cost": "3",
"lead_time": "10",
"max_inventory": "80",
"max_order": "25",
"risk_tolerance": "3"



LLM Consultant

I recommend a continuous review (s, S) policy for inventory management based on the provided parameters.

Policy Recommendation:
- Set a reorder point (s) at 40 units.
- Set an order-up-to level (S) at 65 units.

Explanation and Calculation:

1. **Reorder Point (s):**
The reorder point is determined by the lead time demand and the risk of stockouts. Given a Poisson demand distribution with $\lambda = 8$ units per day and a lead time of 10 days, the expected demand during the lead time is $8 \times 10 = 80$ units. However, to account for daily ordering and max order constraints, we set a more practical reorder point at 40 units considering the max order size and balancing against penalty costs.

2. **Order-up-to Level (S):**
With a maximum inventory constraint of 80 units and to ensure the ability to fulfill orders within this constraint, the order-up-to level is set at 65 units. This provides a buffer against demand variability while staying within the maximum order size of 25 units.

Ordering Policy:
- Monitor the inventory level daily: Calculate the sum of current on-hand inventory and any outstanding orders.
- If this total falls below the reorder point of 40 units, place an order.
- The order quantity should be the difference between the current inventory position (on-hand + on-order) and the order-up-to level (S = 65 units), but not exceeding the max order size of 25 units.

This strategy ensures efficient inventory handling by balancing holding costs and penalty costs while adhering to the maximum inventory and order constraints.